

```
import java.util.Scanner;
```

```
public class Task3 {  
    public static class BankAccount {  
        private double balance;  
  
        public BankAccount(double initialValue) {  
            this.balance = initialValue;  
        }  
  
        public double getBalance() {  
            return balance;  
        }  
  
        public void deposit(double amount) {  
            if (amount > 0) {  
                balance += amount;  
                System.out.println("deposited: £" + amount);  
            } else {  
                System.out.println("enter valid amount");  
            }  
        }  
  
        public void withdraw(double amount) {  
            if (amount > 0 && amount <= balance) {  
                balance -= amount;  
                System.out.println("withdrawn: £" + amount);  
            } else if (amount > balance) {  
                System.out.println("insufficient balance");  
            } else {  
                System.out.println("invalid withdrawal amount");  
            }  
        }  
    }  
}  
  
public static class atm {  
    private BankAccount account;  
    private Scanner reader;  
  
    public atm(BankAccount account) {  
        this.account = account;  
        this.reader = new Scanner(System.in);  
    }  
  
    public void start() {  
        while (true) {  
            showMenu();  
            int options = reader.nextInt();  
            switch (options) {  
                case 1:  
                    checkBalance();  
                    break;  
                case 2:  
                    deposit();  
                    break;  
                case 3:  
                    withdraw();  
                    break;  
                case 4:  
                    System.out.println("thank you, have a nice day.");  
                    return;  
                default:  
                    System.out.println("try again");  
            }  
        }  
    }  
}
```

```
private void showMenu() {
    System.out.println("/nATM menu:");
    System.out.println("1: balance check");
    System.out.println("2: deposit");
    System.out.println("3: withdraw");
    System.out.println("4: exit");
    System.out.print("choose an option: ");
}

private void checkBalance() {
    System.out.println("your current balance is: £" + account.getBalance());
}

private void deposit() {
    System.out.print("enter amount to deposit: ");
    double amount = reader.nextDouble();
    account.deposit(amount);
}

private void withdraw() {
    System.out.print("enter amount to withdraw: ");
    double amount = reader.nextDouble();
    account.withdraw(amount);
}

}

public static void main(String[] args) {
    BankAccount account = new BankAccount(1000);
    atm ATM = new atm(account);
    ATM.start();
}

}
```